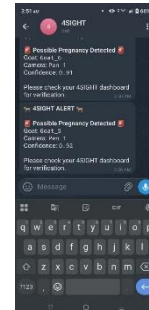


Functions	Types	Proposed Specifications	Actual Specifications	Status
Capture Video	Principal	A camera that can record 1080p footage of goats with Infrared night vision ranging up to 40 feet and has Real Time Streaming Protocol	Goat footage was captured using a TAPO CCTV Camera with RTSP	Done
Store Video Footage	Principal	Store captured video of goats for processing and streaming	Storage of videos was done through a computer for processing	Done
Extract Image	Principal	Deconstruct the incoming video stream into individual frames	Implemented using OpenCV's C++ backend libraries to seamlessly extract continuous frames from the CCTV video feed.	Done
Enhance Image	Secondary	Apply image enhancing algorithms to make identification easier	Integrated the CLAHE algorithm via OpenCV to dynamically adjust contrast and handle low-light/shadowed farm conditions.	Done
Detect Goat	Principal	Utilize object detection to localize and isolate goat subjects within the enhanced frames.	Initialized the YOLOv11 model to draw precise bounding boxes around individual goats within the processed frames.	Done
Track Posture	Principal	Map and monitor skeletal key points and motion patterns of detected goats	Integration of the ByteTrack algorithm mapped directly with YOLOv11 to maintain persistent unique IDs for goats, even during partial occlusions.	Done
Classify Behavior	Principal	Tag behaviors with metadata (timestamp, goat ID) for dataset expansion.	The YOLOv11 and ByteTrack integration successfully appends exact system timestamps and persistent Goat IDs to every recognized behavioral frame for accurate metadata tagging	In progress
		Apply mandatory probability cut-offs (e.g., $\geq 85\%$) to ensure only high-certainty behaviors are recorded.	The model's classification head is programmed with a strict probability threshold; the system successfully filters out low-confidence data, processing and	

			logging only behavioral detections scoring 85% or higher.	
		Categorized movements with specific states (e.g. isolation, resting, aggression, walking, and feeding), timestamp range, and confidence level for identified behaviors.	The unified YOLOv11 model successfully identifies and categorizes some of the target states (isolation, resting, aggression, walking, and feeding), while the backend script records the start/end timestamp ranges and the exact confidence level for each continuous action.	
Log Activity	Principal	Insert and commit detection data (timestamps, Goat IDs, behavioral events, and $\geq 85\%$ confidence scores) into the MySQL database.	Backend database constructed using MySQL to store strictly tabular, timestamped behavioral logs.	Done
Send Notification	Principal	Initiate a real-time automated alert via app notification within 5 seconds of detection, containing the specific behavioral event tag (resting, feeding, isolation, aggression and walking) and a confidence score $\geq 85\%$.	Notification alerts are automatically sent through a Telegram Chatbot API once behavioral events are detected. 	Done
		Automated alert will be sent once the pregnancy behavioral indicators reach certain duration thresholds i.e. (high level of resting, isolation, aggression and low levels of walking, and feeding)	Pregnancy alerts are sent when certain duration thresholds are reached	In Progress

Display Data	Secondary	Render the custom visual dashboard, maintaining a live video feed at a minimum of 30 Frames Per Second (FPS), and automatically refresh the behavioral logs and notification history within < 5.0 seconds of a registered database event.	A locally hosted Custom Web Dashboard (built with HTML/CSS/Python) interfacing directly with the MySQL database and RTSP stream.	Done
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